Vector Spaces and Linear Maps

Course notes - Department of Mathematics

1. Vector spaces

A vector space over a field consists of a set closed under addition and scalar multiplication, satisfying associativity, commutativity of addition, distributivity, and the existence of a zero vector and additive inverses.

A subspace is a subset that is itself a vector space, which reduces to being non-empty and closed under both operations. The span of a set is the smallest subspace containing it.

2. Basis and dimension

A set is linearly independent when no non-trivial linear combination gives the zero vector. A basis is an independent spanning set, and every basis of a given space has the same cardinality, which is the dimension.

Coordinates relative to a basis turn abstract vectors into tuples, which is what makes computation possible. Changing basis is itself a linear map, represented by an invertible matrix.

3. Linear maps and matrices

A linear map preserves addition and scalar multiplication. Once bases are fixed, every linear map between finite-dimensional spaces is represented by a matrix, and composition of maps corresponds to matrix multiplication.

The rank-nullity theorem states that the dimension of the domain equals the rank plus the nullity. It immediately explains when a square system has a unique solution: precisely when the nullity is zero.

4. Determinants

The determinant is the unique alternating multilinear function of the columns taking the value one on the identity. Geometrically it is the signed volume scaling factor of the map.

A matrix is invertible exactly when its determinant is non-zero. Determinants are indispensable in theory but a poor computational tool; Gaussian elimination is what one actually runs.

5. Eigenvalues and diagonalisation

A scalar lambda is an eigenvalue of A when Av = lambda*v for some non-zero v. The eigenvalues are the roots of the characteristic polynomial det(A - lambda*I).

A matrix is diagonalisable when its eigenvectors form a basis, in which case A = PDP^-1 and powers of A become trivial to compute. Symmetric real matrices are always orthogonally diagonalisable with real eigenvalues, which is the spectral theorem.